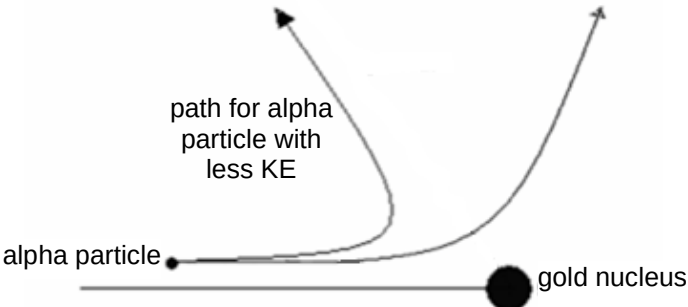
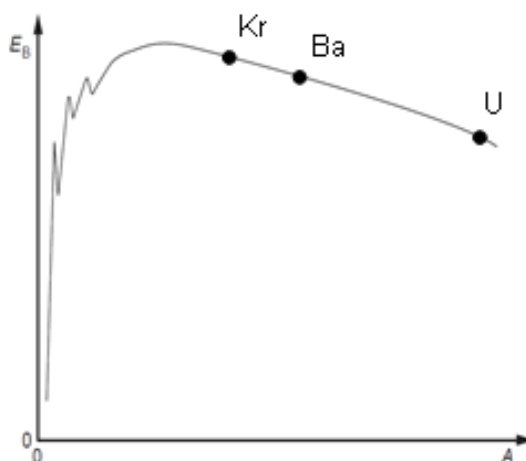


9ai	- nucleus has positive charge, so repels alpha particle. alpha particle gains momentum at right angles to the initial momentum. - greater deflection of final path (path cannot be parallel to original dir) final path has greater distance of closest approach 	B1 B1 B1 B1
9aii	β-particles have a range of energies β-particles deviated by (orbital) electrons / attracted by nucleus β-particle has (very) small mass (any two sensible suggestions, 1 each, max 2)	B1 B1
9bi	- as alpha particle approaches nucleus, KE is converted to EPE, at this distance of closest approach, alpha particle must possess a certain amount of EPE hence alpha particle must possess minimum energy to be this close to Li $E = 1/4\pi\epsilon_0 Q_1 Q_2 / r$ $= 9 \times 10^9 \times 3.2 \times 10^{-19} \times 4.8 \times 10^{-19} / 4.2 \times 10^{-15}$ $= 3.3 \times 10^{-13} \text{ J}$	B1 B1 M1 M1 A0
9bii	mass change = $10.0011 + 1.0087 - 7.0144 - 4.0015 = -6.1 \times 10^{-3} \text{ u}$ mass loss = $6.1 \times 10^{-3} \times 1.66 \times 10^{-27} = 1.01 \times 10^{-29} \text{ kg}$	M1 M1

	2. $E = mc^2$ $= 1.0 \times 10^{-29} \times 9 \times 10^{16} = 9.0 \times 10^{-13} \text{ J}$ Energy of neutron $= 9.0 \times 10^{-13} + 3.3 \times 10^{-13}$ $= 1.2 \times 10^{-12} \text{ J}$	C1 C1 A1
9ci	Refer to summary notes.	B1
9cii	 U labelled near right-hand end of line Ba and Kr in approximate relative positions	B1 B1
9ciii	The binding energy of a particular nucleus $= A \times E_B$ The process is possible because the binding energy of U < the combined binding energy of (Ba + Kr)	B1 B1